



Polarity Tagger

A quick guide for someone who has never used the tool, and has never seen CCG. Understand what it does, learn the grammar, drive the app, then read its output correctly.



PART 1 · THE IDEA

You type a sentence. It labels every word.

Each label is a **polarity**: the direction of valid inference at that position. The whole job is to turn a sentence into a map of where you may reason **up**, **down**, or neither.

PART 1 · THE IDEA

The three tags

↑ up

Replace the word with something **more general**.

"a dog" ↑ gives "an animal"

↓ down

Replace it with something **more specific**.

"no animal" ↓ gives "no dog"

= equal

Neither. The position is **neutral**.

PART 1 · THE IDEA

"no" flips a position from up to down.

no animal moves

⇓ implies

no dog runs

Polarity comes from a deeper property called **monotonicity**, covered in Part 4.

02

The Grammar: CCG

Combinatory Categorical Grammar: how the words snap together into a sentence.

PART 2 · THE GRAMMAR

Almost everything lives in the dictionary.

Each word carries a **category**. A few universal rules combine those categories until the whole sentence becomes one **S**.

Slash notation

X/Y

"I need a **Y** on my **right**; give me one and I become an X."

X\Y

"I need a **Y** on my **left**; give me one and I become an X."

The slot is always written on the right of the slash; the result on the left.

PART 2 · THE GRAMMAR

Reading the lexicon

dog : N	a noun, complete already, needs nothing.
Fido : NP	a noun phrase, complete already.
runs : S\NP	needs an NP on its left (the subject) to become S.
every : NP/N	needs an N on its right to become NP.
chase : (S\NP)/NP	object on the right, then subject on the left.

PART 2 · COMBINATION RULES

Forward application >

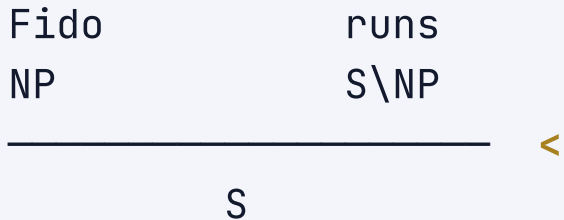
A functor X/Y consumes the Y on its **right**.



PART 2 · COMBINATION RULES

Backward application <

A functor $X \backslash Y$ consumes the Y on its **left**.



PART 2 · COMBINATION RULES

Type-raising $>T$ $<T$

forward $>T$ turns X into $S/(S \backslash X)$

backward $<T$ turns X into $S \backslash (S/X)$

Alone it changes nothing; its job is to **set up composition**. It lets an argument behave like a function, so it can glue to its neighbour before its head arrives.

PART 2 · COMBINATION RULES

Composition $>B$ $<B$ $>B\times$ $<B\times$

$$X/Y + Y/Z \rightarrow X/Z$$

Glue two functors into one without applying yet; crossed variants let the slashes point different ways. Together with type-raising, this is what lets **"John saw"** act as a single constituent, used for relative clauses, extraction, and coordinating mismatched chunks.

PART 2 · COMBINATION RULES

Coordinate Φ

Join two phrases of the **same** category with "and / or / but".

Fido swam	and	Felix ran	
S	CONJ	S	
			Φ
S			

03

Driving the App

The interface runs left to right: **Type** → **Load** → **Build** → **Tag**.

PART 3 · LOAD & BUILD

Load turns text into cards.

every

NP/N

dog

N

runs

S\NP

The tool already knows a fixed set of words. If you would rather not build a CCG derivation yourself, click a ready-made **Example** to load it complete. To see how that example was parsed, press **Undo** to step back through it move by move.

PART 3 · LOAD & BUILD

Word variations are handled for you.

dogs → dog mice → mouse

ran → run saw → see chased → chase

Plurals fold to singular; tensed verbs fold to their base. So "**no dogs ran**" loads the same cards as "**no dog run**".

PART 3 · LOAD & BUILD

Some cards switch ⇄

A dashed underline means more than one reading; click to cycle.

who / whom / that	object extraction (default) or subject extraction
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<u>if</u>	clause-initial "If A, B" or clause-final "B if A"
------------------	---

PART 3 · LOAD & BUILD

Build by combining the cards.

Select **two adjacent** cards, then click **>**, **<**, or a composition.

Select **one** card, then click **>T**, **<T** to type-raise.

Raise a card only when you will **compose** it next. A raised card that is simply applied does not tag.

Select **three adjacent** cards, then click **Φ** to coordinate.

You are done when a single S card remains, ready to Tag.

04

The Semantics: Monotonicity

The tree was pure syntax. Polarity rests on one idea.

PART 4 · MONOTONICITY

Each category gets a semantic type.

These types form a **preordered family** that records monotonicity: every function is increasing, decreasing, or neutral in each argument.

-
- +** **increasing** function: a bigger input gives a bigger output; inference flows the same way.
-
- **decreasing** function: a bigger input gives a smaller output; inference reverses.
-
- **neutral** function: no guaranteed direction either way.

PART 4 · MONOTONICITY

The marks live on the arrows.

some $(e \rightarrow t) \rightarrow + ((e \rightarrow t) \rightarrow + t)$

no $(e \rightarrow t) \rightarrow - ((e \rightarrow t) \rightarrow - t)$

every $(e \rightarrow t) \rightarrow - ((e \rightarrow t) \rightarrow + t)$

most $(e \rightarrow t) \rightarrow \bullet ((e \rightarrow t) \rightarrow + t)$

Read the marks straight off. "no" reverses inference in both arguments, so both arrows are $-$.

"most" says nothing about its noun, so that arrow is $\bullet$.

05

The Typing Rules

Every rule carries a polarity d on its terms and a mark on each arrow.

PART 5 · THE RULES

Polarity **red**, markings **blue**

$$\begin{array}{c}
 \frac{u^{\text{red}d} : \sigma \rightarrow^{\text{blue}m} \tau \quad v^{\text{blue}md} : \sigma}{(uv)^{\text{red}d} : \tau} > \quad \frac{u^{\text{red}d} : \tau \rightarrow^{\text{blue}n} \rho \quad v^{\text{blue}nd} : \sigma \rightarrow^{\text{blue}m} \tau}{(u \circ v)^{\text{red}d} : \sigma \rightarrow^{\text{blue}mn} \rho} \text{ B} \quad \frac{u^{\text{blue}md} : \sigma}{u^{\text{red}d} : (\sigma \rightarrow^{\text{blue}m} \tau) \rightarrow^{\text{blue}+} \tau} \text{ T} \\
 \\
 \frac{u^{\text{blue}md} : e \rightarrow \tau}{u^{\text{red}d} : (n \rightarrow^{\text{blue}m} t) \rightarrow^{\text{blue}+} \tau} \text{ H} \quad \frac{u^{\text{red}d} : \sigma \quad \sigma \leq \sigma'}{u^{\text{red}d} : \sigma'} \text{ W} \quad \frac{u^{\text{blue}=-} : e}{u^{\text{red}d} : np^{\text{blue}+}} \text{ ML} \\
 \\
 \frac{x^{\text{red}d} : \sigma \quad \text{and}^{\text{red}d} : \sigma \rightarrow^{\text{blue}+} (\sigma \rightarrow^{\text{blue}+} \sigma) \quad y^{\text{red}d} : \sigma}{(x \text{ and } y)^{\text{red}d} : \sigma} \Phi
 \end{array}$$

$$n = e \rightarrow t, \quad np^{\text{blue}+} = (n \rightarrow^{\text{blue}+} t), \quad np^{\text{blue}-} = (n \rightarrow^{\text{blue}-} t), \quad np^{\text{blue}\bullet} = (n \rightarrow^{\text{blue}\bullet} t)$$

06

What "Tag" Does

Pressing Tag runs two automatic passes over the finished tree.

Two passes over the tree

PASS 1 ↑

Fit the types

Bottom-up. Each leaf takes its marked type from the lexicon; the tool walks up, checking each step against the rules and computing the result type.

PASS 2 ↓

Push the polarity

Top-down. Start at the root S with **up**; apply the rule at each node until every word is tagged.

A nuance: type-raising is resolved against its neighbour.

A type-raised card carries no fixed type on its own. It must feed a **composition**, either directly or after being coordinated with another type-raised card. A composition whose two sides are **both** type-raised is not supported.

So in Pass 1 the tool does not raise it in the abstract: it looks at the **other premise**, the functor it is composing with, and picks one of two behaviours.

CASE 1 real raise

The card genuinely lifts.

If the card's type matches what the functor expects deep inside, it is raised to a higher type so the two can compose, exactly like the T rule.

CASE 2 identity

Raising is a no-op.

If the card is already shaped to absorb the functor directly, it passes through **unchanged** and the two compose as they are.

Which case applies is a consequence of the **lexicon** you tagged with, itself an input, as the next slide shows. If neither fits, Pass 1 reports that the card cannot be raised against its neighbour.

See the example **every cat who Fido and Felix chased ran**.

PART 6 · TAG

The lexicon is an input too.

The marked types are **editable**, so the result depends on two things you control: the **derivation** you built and the **lexicon** you tagged it with.

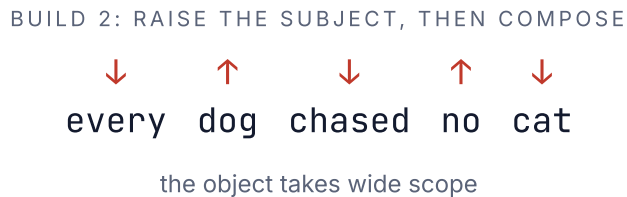
Tag is best-effort with fallbacks. When none applies, Pass 1 **fails** and reports where. The practical goal: get the types to fit.

When Pass 1 fails, the message says **where**, and suggests what to change: add an **Adapt** line, or rebuild that step with **application** instead of composition.

PART 6 · TAG

The derivation is an input too.

One sentence can have **two builds**. Each build gets its own tags, and each set of tags is correct **for that build**.



Only **every** and **dog** change. This is not a bug: each build carries its own **reading**, and the tags are sound for that reading.

PART 6 · TAG

One more: negation and scope.

BUILD 1: APPLY ONLY

= ↑ ↓ ↓ ↓
Fido didn't chase a cat

negation takes wide scope: "Fido chased no cat at all"

BUILD 2: COMPOSE DIDN'T WITH CHASE FIRST

= ↑ ↓ ↑ ↑
Fido didn't chase a cat

the object takes wide scope: "there is a cat Fido didn't chase"

Only **a** and **cat** change. Each build carries its own **reading**, and the tags are sound for that reading. Here Build 1 carries the reading English prefers, and it is the build a parser returns; Build 2 is still correct, for its own reading.

07

The Adapt Function

When a typing step does not fit, Adapt rewrites the two types so the derivation can be typed.

PART 7 · ADAPT

What Adapt does

Adapt is a short list of type adaptations, tried in order. Given a rule and the pair of types it is combining, it returns a new pair of types for that step. While the box is **empty** it behaves as the identity: it returns the input pair unchanged, so the raw algorithm fails exactly where help is needed. A line you add affects only the pairs it matches; every untouched pair passes through as before.

We claim that two adaptations cover almost everything: **H and **W**.**

PART 7 · ADAPT

"no dog runs" gets stuck.

runs has the simple type $e \rightarrow t$, so it expects a plain **entity** as its subject.

But **no dog** is a **quantifier**, a function over properties, not a plain entity.

The types do not fit, so Pass 1 stops here. This is where **H** comes in.

PART 7 · ADAPT

H: verb-lifting

H enlarges the verb's argument so it can take a quantifier instead of a bare entity.

In the tool, H is **two lines**, one for each result type of the verb: **t** for an intransitive verb, **e→t** for a transitive one.

RULE

$$\frac{u^{md} : e \rightarrow \tau}{u^d : (n \rightarrow^m t) \rightarrow^+ \tau} \text{ H}$$

EXAMPLE: NO DOG RUNS

$$\frac{\frac{no\ dog^\uparrow : n \rightarrow^- t \quad \frac{runs^\downarrow : e \rightarrow t}{runs^\uparrow : (n \rightarrow^- t) \rightarrow^+ t} \text{ H}}{no\ dog\ runs^\uparrow : t} <$$

PART 7 · ADAPT

W: weakening for coordination

When two conjuncts have different types, W sends both to their **join**: keep **+** where both agree, **-** where both agree, **•** elsewhere.

RULE

$$\frac{u^d : \sigma \quad \sigma \leq \sigma'}{u^d : \sigma'} \text{ W}$$

EXAMPLE: SOME DOG AND NO CAT

$$\frac{\frac{\text{some dog} : np^+}{\text{some dog} : np^\bullet} \text{ W} \quad \text{and} : np^\bullet \rightarrow^+ (np^\bullet \rightarrow^+ np^\bullet) \quad \frac{\text{no cat} : np^-}{\text{no cat} : np^\bullet} \text{ W}}{\text{some dog and no cat} : np^\bullet} \Phi$$

PART 7 · THE PRACTICAL LOOP

Build the tree



Tag



does not fit? add H or W



Tag again

Usually H and W are enough. If they are not, and you find yourself adding a new condition to the Adapt function, please share it with us so we can improve the program.

Writing your own Adapt line

A new line brings three things: the **pair of types** it rewrites, a **marking** for the child, and the **operation** it performs on the meaning. Then there is **one check**:

marking	the check on the operation
+	it must be increasing : larger inputs give larger outputs
-	it must be decreasing : larger inputs give smaller outputs
•	nothing to check

08

Reading the Output

Every word carries a final tag. Read each as a substitution license.

PART 8 · READING THE OUTPUT

Each tag is a substitution license.

↑ **up** replace with something **more general**.

↓ **down** replace with something **more specific**.

= **equal** neither substitution is licensed here.

PART 8 · READING THE OUTPUT

"no dog runs", tagged

↑ ↓ ↓
no dog runs

Under "no", both "dog" and "runs" come out **down**, so you may only specialise them. That is exactly why **"no animal moves" implies "no dog runs"**, and **"no dog runs" implies "no pug sprints"**.

PART 8 · READING THE OUTPUT

Change the determiner, change the polarity

Sentence	dog	runs	because the determiner is...
some dog runs	↑ up	↑ up	increasing in both arguments
no dog runs	↓ down	↓ down	decreasing in both arguments
every dog runs	↓ down	↑ up	decreasing noun, increasing verb
most dogs run	= equal	↑ up	neutral noun, increasing verb

The same phrase, built two ways

COMPOSITION

Fido didn't chase Felix

didn't is first composed with the verb, then the result applies to the object.

APPLICATION *normal form*

Fido didn't chase a cat

didn't is applied to the whole verb phrase, once the phrase is complete.

Two examples build the same verb phrase in different ways. Only the second is a **normal-form** derivation. The tool accepts both: it tags the derivation you give it, and the output describes the **reading** that this derivation carries.

A proper name hides the difference. With a quantifier it shows: the earlier **negation and scope** slide builds **Fido didn't chase a cat** both ways, and the tags on **a** and **cat** change.

Now try it.

Type a sentence → Build to one S → Tag → H / W if needed → Read ↑ ↓ =

Or load an Example and Undo through it to see how a finished derivation was built.